

PASAN DISSANAYAKE

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RESEARCH INTERESTS

My research focuses on efficient and sustainable machine learning, with an emphasis on model reconstruction and knowledge distillation techniques to democratize reliable AI. I aim to extract and transfer large model intelligence into compact architectures by systematically integrating interpretability/explainability methods. My research provides theoretical guarantees, leveraging optimization, information theory, and statistics, along with experimental validation on real-world datasets and benchmarks.

EDUCATION

University of Maryland, College Park, MD Started Aug. 2022
PhD in Electrical and Computer Engineering (CGPA: 3.97/4.0)
Advised by Prof. Sanghamitra Dutta

University of Moratuwa, Sri Lanka Graduated Aug. 2021
BSc (Hons.) in Electronic and Telecommunication Engineering (GPA: 4.04/4.2)
Advised by Dr. Prathapasinghe Dharmawansa, Dr. Ranga Rodrigo, and Dr. Peshala Jayasekara
Thesis: Precise vehicle localization using fusion of multiple sensors for self-driving

PUBLICATIONS

Conference:

- **P. Dissanayake**, P. -Y. Chen, S. Dutta, “Accelerating Multi-Property Molecular Design via Entropic-Risk-Based Counterfactual Explanations,” International Conference on Learning Representations (ICLR) workshop on AI for Accelerated Materials Design, 2026.
- **P. Dissanayake**, F. Hamman, B. Halder, I. Sucholutsky, Q. Zhang, S. Dutta, “Quantifying Knowledge Distillation using Partial Information Decomposition,” Artificial Intelligence and Statistics (AISTATS), 2025
- F. Hamman, **P. Dissanayake**, Y. Fu, S. Dutta, “Few-Shot Knowledge Distillation of LLMs With Counterfactual Explanations,” Neural Information Processing Systems (NeurIPS), 2025
- F. Hamman, **P. Dissanayake**, S. Mishra, F. Lecue, S. Dutta, “Quantifying Prediction Consistency Under Model Multiplicity in Tabular LLMs,” International Conference on Machine Learning (ICML), 2025
- S. Meel, **P. Dissanayake**, M. Nomeir, S. Dutta, S. Ulukus, “Private Counterfactual Retrieval With Immutable Features,” IEEE International Symposium on Information Theory (ISIT), 2025
- E. Noorani, **P. Dissanayake**, F. Hamman, S. Dutta, “Counterfactual Explanations for Model Ensembles Using Entropic Risk Measures,” Autonomous Agents and Multiagent Systems (AAMAS), 2025.
- M. Nomeir, **P. Dissanayake**, S. Meel, S. Dutta, S. Ulukus, “Private Counterfactual Retrieval,” Allerton Conference on Communication, Control, and Computing, 2025
- **P. Dissanayake** and S. Dutta, “Model Reconstruction Using Counterfactual Explanations: A Perspective From Polytope Theory,” Neural Information Processing Systems (NeurIPS), 2024

Journal:

- B. Halder, F. Hamman, **P. Dissanayake**, Q. Zhang, I. Sucholutsky, S. Dutta, “Quantifying Spuriousness of Biased Datasets Using Partial Information Decomposition,” Transactions on Machine Learning Research (TMLR), 2025
- **P. Dissanayake**, P. Dharmawansa and Y. Chen, “Distribution of the Scaled Condition Number of Single-Spiked Complex Wishart Matrices,” in IEEE Transactions on Information Theory, vol. 68, no. 10, pp. 6716-6737, 2022
- P. Dharmawansa, **P. Dissanayake** and Y. Chen, “The Eigenvectors of Single-Spiked Complex Wishart Matrices: Finite and Asymptotic Analyses,” in IEEE Transactions on Information Theory, vol. 68, no. 12, 2022
- **P. Dissanayake** and S. Dutta, “Foundations of Efficient Model Reconstruction Using Counterfactual Explanations,” under review.
- S. Meel, M. Nomeir, **P. Dissanayake**, S. Dutta, S. Ulukus, “What If, But Privately: Private Counterfactual Retrieval,” under review.

RESEARCH PROJECTS

Accelerated Material Discovery Using Counterfactual Explanations (*On-going*)

The goal is to identify new biopolymer nanocomposites as sustainable plastic alternatives that satisfy multiple desirable properties simultaneously. Developed a new inverse design algorithm that generates counterfactuals for multiple targets within seconds as compared to brute force which could take hours.

Extreme distillation: Few-Shot Distillation of LLMs into MLPs (*On-going*)

Investigating the feasibility of distilling large language models (LLMs) into task-specific multi-layer perceptrons (MLPs) within a meta-learning setting. The goal is to yield the world knowledge of the LLM for inferring parameters for classical ML models with only a few shots. Experiments are conducted on real-world tabular datasets.

Quantifying task-related knowledge in knowledge distillation

Introduced task-aware metrics to measure knowledge available to distill and already distilled information based on Partial Information Decomposition (PID). Implemented alternating-optimization algorithm on CIFAR-10/100 and CUB-200-2011 datasets using PyTorch (on WideResNet and ResNet models). [AISTATS 2025]

Counterfactual explanations and model reconstruction. Investigated the use of counterfactual explanations for model reconstruction (i.e., explanations only from one side of the decision boundary). Implemented algorithm on tabular datasets using TensorFlow. Extended to LLMs. Also, developed private retrieval mechanisms for counterfactuals within a PIR setting. [NeurIPS 2024, ISIT 2025, AAMAS 2025, Allerton 2025, NeurIPS 2025]

Quantifying prediction consistency of tabular LLMs under fine-tuning multiplicity. Developed a metric to efficiently quantify model multiplicity of LLMs fine-tuned on tabular datasets. Experiments were done with BigScience T0 and Google FLAN T5 models from Huggingface. LORA and T-Few methods were used as the fine-tuning techniques. [ICML 2025]

Single-spiked complex Wishart matrices. Studied the properties of two important statistics of the complex Wishart ensemble, with a focus on the centralized (zero-mean) case with a single-spiked covariance structure. Simulations were done using Mathematica. [IEEE Trans. Info. Theory, 2022]

AWARDS AND HONORS

- Outstanding Graduate Assistant (2024) - Graduate School, University of Maryland
- Outstanding Teaching Assistant (2025, 2024) - ECE, University of Maryland
- Dean's Fellowship (2022) - University of Maryland
- Dean's List (in all 8 semesters) - University of Moratuwa
- Google Hash Code (2019) Country rank: 3

CERTIFICATIONS

- Reinforcement Fine-Tuning LLMs With GRPO - Predibase on DeepLearning.ai
- Fundamentals of AI Agents Using RAG and LangChain - IBM on Coursera
- Machine Learning - Stanford University on Coursera
- Data Structures, Algorithmic Toolbox - University of California, San Diego on Coursera
- Certificate in Software Development in Java - National Institute of Business Management, SL
- Networking Essentials (NE-CS3032-IN-16-EN) - Offered by Cisco Networking Academy

EMPLOYMENT

Lecturer (on Contract) Dec. 2021 to Aug. 2022
Department of Electronic and Telecommunication Engineering,
University of Moratuwa, Sri Lanka

Systems Engineer (Intern) Jun. 2019 to Dec. 2019
Infrastructure Solutions Team
MillenniumIT ESP (Pvt) Ltd., Colombo 07, Sri Lanka

TEACHING EXPERIENCE

University of Maryland, College Park, MD ENEE621: Estimation and Detection Theory (Spring 2025); ENEE324: Engineering Probability (Spring 2023, Fall 2023, Spring 2025); ENEE436: Foundations of Machine Learning (Fall 2023); ENEE420: Communication Systems (Fall 2022)

University of Moratuwa, Sri Lanka EN4720: Security in Cyber-Physical Systems; EN2550: Fundamentals of Image Processing and Machine Vision; EN3053: Digital Communications - I

SERVICES AND EXTRA-CURRICULAR

- Reviewer - ICML 2026, NeurIPS 2025; ICLR 2025; XAIFIN 2024; ISIT 2025, 2024; FAccT 2025, 2024; IEEE Wireless Communications Letters (*view on Publons*)
- Project Lead - UMD Trustworthy AI in Law and Society (TRAILS) AI Summer Camp 2024
- Member of the panel of judges - UMD Louis Stokes Alliance for Minority Participation (LSAMP) Fall Research Symposium 2023
- Mentor - Montgomery Blair High School Magnet Student Program 2023
- Secretary - Drama Society, University of Moratuwa for the year 2020/2021